

Call & Response MIDI — User Manual

(v2.0.0)

Viral Samples | Intelligent Algorithmic Phrase Generator & 303-Style Sequencer

Formats: VST3, Audio Unit (AU), Standalone / Platforms: macOS (Universal: Mojave Intel & Apple Silicon) & Windows (64-bit)

1. Overview & Architecture

Call & Response MIDI v2.0.0 is an advanced dual-lane algorithmic melody generator, performance sequencer, and MIDI composition instrument inspired by classic Roland TB-303 phrasing and the timeless musical dialogue of *Call & Response*.

Instead of disconnected random notes, Call & Response MIDI constructs coherent pairs of musical phrases that converse with one another:

- **Call Lane:** The theme, question, or melodic hook.
- **Response Lane:** The counter-melody, answer, or harmonic resolution shaped by sophisticated modal and rhythmic response strategies.

Version 2.0.0 introduces a completely redesigned, professional dark Ableton-inspired interface with fixed aspect-ratio scaling, interactive step-and-dropdown ribbon controls, continuous velocity and swing dragging, independent lane muting, lane locking, lane mutation percentage controls, rectangular note offset shifters, and an integrated 4-tab Inspector.

A built-in analog-modelled 303 preview synthesizer with resonant 24 dB low-pass filter and snappy envelope decay enables immediate auditioning without needing to route to an external synth.

2. What's New in Version 2.0.0

- **Sleek Modern UI:** Fully overhauled dark-mode interface with refined typography, high-contrast visual cues, and fixed proportional aspect-ratio scaling (`1180 \times 750`).

- **Top Ribbon Controls:**
- Direct value displays for **OCT**, **KEY**, **SCALE**, **b/#**, **STEPS**, **RATE**, and **SWING**.
- Double interaction paradigm: click to open dropdown selection or click the **▲** / **▼** arrow steppers.
- **Tempo-Relative RATE:** Switch between **Half speed**, **0** (Normal tempo sync), and **Double speed**.
- **Continuous SWING Dragging:** Click and drag vertically to smoothly adjust groove swing from **0%** to **100%**.
- **8 Rapid-Access Preset Slots (A–H):** Instant **Cmd+Click** saving, **Double-Click** loading, and right-click context menus.
- **Dedicated Undo & Redo:** Fast step-by-step history undo/redo for parameters and note adjustments.
- **Global PANIC Button:** Instant MIDI All-Notes-Off reset across all channels.
- **PREVIEW Audio Toggle:** One-click internal 303 synth enable/disable.
- **Left Control Panel:**
- High-visibility **NEW PAIR** and **DICE** buttons.
- Master **DICE** randomizes unlocked Style, Strategy, Motif, Mode, and Ending settings.
- Rotary knobs for **DENSITY**, **MUTATE**, and **OCT JUMP**.
- Rectangular **[<] OFFSET [>]** buttons for nudging note patterns left and right.
- **Dual Phrase Lanes (CALL & RESPONSE):**
- **Color-Coded Tracks:** Vibrant orange for Call and deep cyan/teal for Response.
- **MUTE Buttons:** Cleanly bypass sound and playback independently for each lane.
- **DICE Mutator:** Rerolls content specifically for Call or Response.
- **Modifiable MUTATE %:** Clickable percentage value to set custom per-lane mutation depth.
- **Lane Lock (Padlock):** Protects either lane from being regenerated during global rolls.
- **Octave Transposition:** Individual **+** / **-** octave buttons per lane.
- **Quick Tools:** Invert, Reverse, Copy, and dedicated MIDI channel routing displays.
- **4-Tab Inspector Panel:**
- **GENERATE:** Style (40 styles), Motif (24 types), Strategy (38 types), Mode (Call then response, Overlapped, Call solo), Ending (Resolve, Unresolved, Surprise) with individual vector lock padlocks.

- **SHAPE:** Gate length (0.25×2.0), continuous click-and-drag **VELOCITY** ($1-127$), Resolve, and Drive.
 - **ARTICULATION:** Accent Boost, Slide Time, Filter Cutoff ($60\text{ Hz}-14\text{ kHz}$), Resonance, and Decay Time.
 - **OPTIONS:** Routing subtab (MIDI Channels 1–16, Combined / Split / Layered) and Timing subtab (Sync Mode: Timeline, Restart, Free).
 - **Interactive Piano Roll:**
 - Matching lane note colors (Orange for Call, Teal for Response).
 - Repeatable right-edge gate resizing up to 4.0 length.
 - Step auditioning with automatic sound stop on mouse release.
 - 2D pitch transposition, time shifting, and double-click note creation/deletion.
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3. System Requirements & Compatibility

3.1 macOS Compatibility

- **Operating System:** macOS 10.14 (Mojave) or later (Catalina, Big Sur, Monterey, Ventura, Sonoma, Sequoia fully verified).
- **CPU Architectures:**
 - **Apple Silicon Native** (`arm64`): M1, M2, M3, M4, Pro, Max, Ultra).
 - **Intel Native** (`x86_64`): Core i5, i7, i9, Xeon).
- **Plugin Formats:** VST3 (64-bit Universal), Audio Unit / AU (64-bit Universal), Standalone Application.
- **AU Validation:** 100% passed Apple `auval` tool.

3.2 Windows Compatibility

- **Operating System:** Windows 10 or Windows 11 (64-bit).
- **CPU Architectures:** 64-bit x86-64 (Intel Core / AMD Ryzen).
- **Plugin Formats:** VST3 (64-bit), Standalone Application (`.exe`).
- **Dependencies:** Completely self-contained with static C++ runtime linking (no external MinGW DLLs required).

3.3 DAW Compatibility

- Ableton Live 10, 11, 12 (macOS & Windows)
 - Logic Pro X & GarageBand (macOS Audio Unit)
 - FL Studio 20, 21, 24 (macOS & Windows)
 - Cubase 11, 12, 13, 14 & Nuendo (macOS & Windows)
 - Studio One 5, 6, 7 (macOS & Windows)
 - Reaper 6, 7 (macOS & Windows)
 - Bitwig Studio 4, 5 (macOS & Windows)
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4. Interface Tour & Controls

4.1 Top Ribbon Bar

[CALL & RESPONSE v.2] OCT: 3 KEY: C SCALE: Minor b/# STEPS: 16 RATE: 0

1. **Plugin Title** Displays **CALL & RESPONSE v.2** and developer attribution **Viral Samples**.
2. **OCT (Octave)**: Baseline pitch octave (**C1** – **C5**). Change by clicking the box for a dropdown or using **▲**/**▼**.
3. **KEY**: Musical root note (**C** through **B**). Supports flats and sharps via the **b/#** toggle button.
4. **SCALE**: 35 curated musical scales. When changed, existing notes intelligently conform to the new scale while preserving interval relationships.
5. **STEPS**: Phrase loop length: **4**, **8**, **12**, or **16** steps.
6. **RATE** Tempo-relative playback speed:
 - **Half speed**: Runs at half the host DAW tempo (**0.5\times**).
 - **0**: Standard **1\times** tempo sync (**1/16\text{th}** notes at host tempo).
 - **Double speed**: Runs at double the host DAW tempo (**2.0\times**).
7. **SWING**: Groove swing (**0\%-100\%**). Click and drag vertically for continuous adjustments, or click the stepper buttons.

8. Preset Slots [A]–[H]

- **Cmd+Click** (or **Ctrl+Click** / **Shift+Click** / **Alt+Click**): Save current pattern and settings to the slot.
- **Double-Click**: Load pattern and settings from the slot.
- **Right-Click**: Open context menu to Save or Load.
- Active slots display a bright cyan indicator badge.

9. **UNDO / REDO**: Instantly revert or restore note edits, generator rolls, and parameter changes.

10. **PANIC**: Sends All-Notes-Off and All-Sound-Off across all 16 MIDI channels.

11. **PREVIEW: ON / OFF**: Enables or mutes the internal 303 synthesizer engine.

4.2 Left Control Section

1. **NEW PAIR**: Generates a brand-new correlated Call and Response melody using the active settings.

2. **DICE**: Rerolls unlocked parameters (Style, Strategy, Motif, Mode, Ending) and regenerates the phrases.

3. **Performance Knobs**:

- **DENSITY**: Adjusts the rhythmic population of notes without disrupting key downbeats.
- **MUTATE**: Governs mutation intensity applied during generative variations.
- **OCT JUMP**: Injects classic 303-style octave leaps (**\pm 12**, **+24** semitones).

4. **[<] OFFSET [>]**: Rectangular buttons that shift all notes in the active phrases left or right by one step.

4.3 Dual Phrase Lanes (CALL & RESPONSE)

Each lane controls its phrase independently:

- **MUTE**: Silences the lane during playback and stops MIDI output for that voice.
- **DICE**: Generates a musical variation specifically for that lane.
- **MUTATE xx%**: Modifiable mutation percentage depth for surgical phrase variations.

- **LOCK (Padlock):** Protects the lane from being altered when pressing NEW PAIR or master DICE.
 - **OCT (+/-):** Transposes the lane up or down by 12 semitones.
 - **INV:** Inverts pitch contours across the active scale.
 - **REV:** Reverses the phrase rhythmically from back to front.
 - **COPY:** Clones the current lane's phrase into the opposite lane.
 - **CH (Channel):** Displays and routes the lane's MIDI output channel (e.g. **CH: 1** or **CH: 2**).
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4.4 Four-Tab Inspector Panel

Tab 1: GENERATE

- **Style (40 Musical Styles):** Governs syncopation, rhythmic density, and slide/accent character (e.g., *Classic Acid*, *Raw Acid*, *Squelchy Acid*, *Dark Rave*, *Driving Techno*, *Minimal Techno*, *Jackin House*, *Deep House*).
- **Motif (24 Hook Types):** Determines the lead-in pattern (*Pulse*, *Offbeat*, *Syncopated*, *Rolling*, *Acid nibble*, *Arp fragment*).
- **Strategy (38 Response Types):** Controls how the Response answers the Call (*Same rhythm + resolve*, *Invert contour*, *Answer downward*, *Mirror contour*, *Echo*, *Response resolves to root*).
- **Mode:** Call then response, Overlapped, or Call solo.
- **Ending:** Resolve to tonic, Unresolved tension, or Surprise cadence.
- **Vector Lock Padlocks:** Click any padlock icon to freeze that parameter during master DICE re-rolls.

Tab 2: SHAPE

- **Gate Length:** Global note length multiplier (**0.25\times** staccato to **2.0\times** legato).
- **Velocity:** Click and drag vertically to smoothly set note velocity between **1** and **127**.
- **Resolve:** Strength of harmonic pull toward tonic root notes on cadence points.
- **Drive:** Saturation drive applied to the internal 303 synth filter stage.

Tab 3: ARTICULATION

- **Accent Boost:** Volume and filter envelope boost applied to accented steps.
- **Slide Time:** Portamento glide time between tied or slurred notes.

- **Filter Cutoff:** 303 low-pass filter frequency (Hz to kHz).
- **Resonance:** Peak resonance emphasizing classic squelchy acid harmonics.
- **Decay Time:** Envelope decay duration (s to s).

Tab 4: OPTIONS

- **Routing Subtab:**
 - *Combined:* Both Call and Response output on MIDI Channel 1.
 - *Split Channels:* Call routes to Channel 1, Response routes to Channel 2 for multi-timbral setups.
 - *Separate:* Independent channel assignments for Call and Response.
 - *Layered:* Both phrases broadcast across multiple selected channels.
 - **Timing & Clips Subtab:**
 - *Sync Mode:* Timeline (synced to DAW transport bar position), Restart (restarts on DAW transport play), Free (runs continuously).
 - *Clip Quantization:* Set re-roll and regeneration quantization to 1 Bar, 1/2 Bar, 1/4 Bar, or Immediate.
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4.5 Interactive Piano Roll

- **Visual Clarity:** Call notes appear in vibrant orange; Response notes appear in bright cyan/teal.
 - **Add / Delete Notes:** Double-click any empty grid cell to add a note; double-click an existing note to delete it.
 - **Move Notes:** Click and drag any note horizontally to shift steps or vertically to transpose pitch.
 - **Resize Gate Duration:** Hover over the right edge of any note (the cursor transforms to ) and drag horizontally. Notes can be lengthened or shortened repeatedly up to times.
 - **Audition Steps:** Click on any note to audition its sound through the internal synth. Auditioning automatically ceases immediately upon mouse release.
 - **Piano Keyboard:** Click the piano keys on the left margin to audition individual pitches.
 - **Accent & Slide:** Right-click a note to toggle Accent (marked with ); Alt-click a note to toggle Slide (marked with ).
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5. MIDI Drag & Drop Export

1. Locate the **DRAG MIDI** button at the bottom of the interface.
 2. Click and hold the **DRAG MIDI** icon.
 3. Drag directly onto any MIDI track in your DAW arrangement (Ableton Live, FL Studio, Logic Pro, Studio One, etc.).
 4. The generated MIDI file includes:
 - Fully resolved note pitches matching your key and scale.
 - Precise note start times, swing offsets, and gate durations.
 - Accurate note velocities with accent boosts.
 - Separate MIDI channels if Split Routing is enabled.
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6. Preset Management & Storage

Presets are stored in standard cross-platform JSON format:

- **macOS:** `~/Library/Application Support/Viral Samples/Call & Response MIDI/Presets/`
- **Windows:** `%APPDATA%\Viral Samples\Call & Response MIDI\Presets\`

Each preset saves:

- All 16 steps of both Call and Response phrases (pitch, length, velocity, accent, slide).
 - Key, Scale, Root Octave, Steps, Rate, and Swing values.
 - Selected Style, Motif, Strategy, Mode, and Ending.
 - All Shape, Articulation, and Routing options.
 - State of all parameter locks and lane mute/lock statuses.
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7. Troubleshooting & FAQ

Q: The plugin does not appear in my DAW on macOS.

A: Ensure you ran the official `.pkg` installer. On macOS, run `killall -9 AudioComponentRegistrar` in Terminal or restart your Mac to refresh the AU plugin cache.

Q: How do I mute the internal synth and use external instruments?

A: Click **PREVIEW: ON** in the top ribbon so it toggles to **PREVIEW: MUTED**. Route the MIDI output of the Call & Response track to your desired software synth track in your DAW.

Q: My DAW's tempo changes, how does RATE respond?

A: Call & Response MIDI automatically locks to host tempo. When **RATE** is set to **0**, phrases run in standard `1/16\text{th}` note sync. Set to **Half speed** for half-time grooves or **Double speed** for drum & bass / double-time arpeggios.

Q: How do I prevent my custom Call phrase from changing when clicking DICE?

A: Click the **LOCK** padlock on the CALL lane header. The Call melody will remain completely intact while the master DICE or lane DICE rolls the Response.

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